

QUESTIONS

Q1

Student management system
10 marks

Q2

Course Enrolment system
10 marks

2/2 answered

Q1 Student management system

10 marks

You are building a student management system where you need to maintain an ordered list of students as they register. Students can register, drop out, or change their position in the waiting list.

perform the following operations

1. create 2 list waiting list and confirmed list
2. get a dropout student name from keyboard - search is it in the waiting list (list contains "Ramesh")
3. if so remove from the list
4. sort the confirmed students list
5. merge waiting list with confirmed list

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         ArrayList<String>waiting=new ArrayList<>();
6         waiting.add("harshi");
7         waiting.add("sneha");
8         waiting.add("balasree");
9         ArrayList<String>confirmed=new ArrayList<>();
10        confirmed.add("vasudha");
11        confirmed.add("bhavya");
12        confirmed.add("paddhu");
13        String name=sc.nextLine();
14        ArrayList<String>newWaiting=new ArrayList<>();
15        for(String s:waiting){
16            if(!s.equalsIgnoreCase(name)){
17                newWaiting.add(s);
18            }
19        }
20        Collections.sort(confirmed);
21        newWaiting.addAll(confirmed);
22        System.out.println(newWaiting);
23    }
24 }
```

Input

stdin input

Output

```
[sneha, balasree, bhavya, paddhu,
vasudha]
```

QUESTIONS

Q1

Student management system
10 marks

Q2

Course Enrolment system
10 marks

2/2 answered

Q2 Course Enrolment system

10 marks

You are developing a course enrollment system where each course can have multiple students, but each student can enroll in a course only once. The system needs to prevent duplicate enrollments and perform various set operations.

perform the following

1. create 2 courses javacourse and python course
2. add 5 students in each
3. check student "rama" is there, if so remove from the list
4. find the union, intersection of both list
5. create treeset, linkedhashset to maintain order and alphabetical order

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         HashSet<String>javaCourse=new HashSet<>();
5         javaCourse.add("rama");
6         javaCourse.add("arun");
7         javaCourse.add("bala");
8         javaCourse.add("kumar");
9         javaCourse.add("sita");
10        HashSet<String>pythonCourse=new HashSet<>();
11        pythonCourse.add("rama");
12        pythonCourse.add("anu");
13        pythonCourse.add("bala");
14        pythonCourse.add("meena");
15        pythonCourse.add("ravi");
16        HashSet<String>temp=new HashSet<>();
17        for(String s:javaCourse){
18            if(!s.equals("rama")){
19                temp.add(s);
20            }
21        }
22        javaCourse=temp;
23        HashSet<String>union=new HashSet<>(javaCourse);
```

Input

stdin input

Output

```
Java Course: [sita, arun, bala,
kumar]
Python Course: [rama, ravi, anu,
bala, meena]
Union: [sita, rama, arun, ravi,
anu, bala, kumar, meena]
```